



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: HTTP Analysis Using Wireshark: Text

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Course Title: Basic Networking Skills for Digital Forensics

Course Code: SBT-DF203

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EXECUTIVE SUMMARY

This report documents the execution and forensic findings of **Lab 3: SYN Flood Attack Investigation Using tshark**. The primary objective was to establish a network traffic baseline for a complete TCP three-way handshake, execute a bounded 4-packet SYN probe on the loopback interface using Python Scapy, and extract diagnostic network indicators using tshark.

All practical activities were conducted inside an isolated Kali Linux virtual machine running Apache2 on 127.0.0.1:80. Analysis verified that while standard baseline traffic achieves full handshake completion (SYN → SYN-ACK → ACK), the bounded probe produced half-open connection attempts (SYN without final ACK). SHA-256 cryptographic hashes verified byte-for-byte evidence integrity across working copies.

SECTION A: THEORETICAL OVERVIEW & CORE CONCEPTS

1. The TCP Three-Way Handshake vs. Half-Open SYN Flooding

Reliable communication over TCP requires an initial three-step exchange before application-layer data can be transmitted:

1. **SYN (Synchronize):** The client sends a packet with the SYN flag set (`tcp.flags.syn==1 && tcp.flags.ack==0`) to request a connection.
2. **SYN-ACK (Synchronize-Acknowledge):** The server responds with SYN and ACK flags set (`tcp.flags.syn==1 && tcp.flags.ack==1`), allocating memory in its connection backlog queue.
3. **ACK (Acknowledge):** The client returns an ACK flag (`tcp.flags.ack==1`) to complete the handshake and open the session.

In a **SYN Flood Attack**, an attacker sends a high volume of SYN requests—often with spoofed or randomized source ports—and deliberately withholds the final ACK. This leaves server sockets in a half-open state until connection timeouts expire, rapidly exhausting TCP backlog queues and causing Denial of Service (DoS) for legitimate users.

2. Forensic Indicators in SYN Activity

A forensic investigator differentiates normal web traffic from SYN probes using specific tshark display filters and metrics:

- **High SYN-to-ACK Ratio:** Anomalous disproportion of initial SYN requests relative to completed handshakes.
- **Ephemeral Port Randomization:** Rapid connection attempts originating from sequentially or randomly generated client source ports (tcp.srcport).
- **Absence of Application Payloads:** Connection attempts that terminate without sending HTTP requests or higher-layer protocol data.

3. Evidentiary Integrity and Cryptographic Hashing

To maintain chain-of-custody standards in digital forensics, primary packet capture files (.pcap / .pcapng) are stored read-only in an evidence/ directory. Working copies are duplicated to a working/ directory using timestamp preservation flags (cp --preserve=timestamps). Pre- and post-analysis verification via SHA-256 ensures that forensic processing does not alter original evidence.

SECTION B: PRACTICAL LAB EXECUTION & EVIDENCE ANALYSIS

TASK 1: ENVIRONMENT PREPARATION & DIRECTORY STRUCTURE

A structured directory layout was created under ~/SBT-DF203-Lab3 to isolate evidence, analysis outputs, exported objects, and reports.

Bash

```
mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}
```

```
cd ~/SBT-DF203-Lab3
```

```
sudo apt update && sudo apt install -y apache2 tshark wireshark python3-scapy
```

```
sudo systemctl enable --now apache2
```

[INSERT SCREENSHOT HERE: Screenshot (358).jpg]

Figure 1: Terminal output showing directory initialization (mkdir -p) and structure validation (find -maxdepth 1).

Optional Capture Acquisition & Baseline Hashing:

Bash

```
wget -O evidence/mySYNFloodCapture.pcap
'https://raw.githubusercontent.com/frankwxu/digital-forensics-
lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'

sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
```



```
(garba@kali)~[~/SBT-DF203-Lab3]
$ sudo systemctl enable --now apache2

(garba@kali)~[~/SBT-DF203-Lab3]
$ wget -O evidence/mySYNFloodCapture.pcap \
>
(garba@kali)~[~/SBT-DF203-Lab3]
$ 'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
zsh: no such file or directory: https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap

(garba@kali)~[~/SBT-DF203-Lab3]
$ wget -O evidence/mySYNFloodCapture.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
--2026-09-11 21:20:03-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
^[[A^c

(garba@kali)~[~/SBT-DF203-Lab3]
$ sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/mySYNFloodCapture.pcap

(garba@kali)~[~/SBT-DF203-Lab3]
```

Download of supplied evidence file mySYNFloodCapture.pcap and initial SHA-256 baseline hashing.

TASK 2: PART A – ESTABLISH NORMAL HANDSHAKE BASELINE

A baseline HTTP capture was recorded on the loopback interface (lo) while initiating a non-persistent HTTP request via curl.

Bash

```
sudo tshark -i lo 'tcp port 80' -a duration:15 -w evidence/normal_http.pcapng &
sleep 2
curl --no-keepalive http://127.0.0.1/ >/dev/null
wait
tshark -r evidence/normal_http.pcapng -Y 'tcp.flags.syn==1 || tcp.flags.fin==1' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/normal_handshake_flags.tsv
```

```

(garba@kali) [~/SBT-DF203-Lab3]
$ sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/normal_http.pcapng &
sleep 2
curl --no-keepalive http://127.0.0.1/ >/dev/null
wait

sudo tshark -r evidence/normal_http.pcapng -Y 'tcp.flags.syn==1 || tcp.flags.fin==1' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/normal_handshake_flags.tsv

[1] 7227
Running as user "root" and group "root". This could be dangerous.
tshark: Couldn't load plugin 'falco-events.so': /usr/lib/x86_64-linux-gnu/libgrpc++.so.1.51: undefined symbol: _Z25absl_status_to_grpc_errorN4absl8debian106StatusE
Capturing on 'Loopback: lo'
tshark: The file to which the capture would be saved ("evidence/normal_http.pcapng") could not be opened: Permission denied.
[1] + exit 1 sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w
% Total % Received % Xferd Average Speed Time Time Current
Dload Upload Total Spent Left Speed
100 10703 100 10703 0 0 2.15M 0 0 0 0
Running as user "root" and group "root". This could be dangerous.
tshark: Couldn't load plugin 'falco-events.so': /usr/lib/x86_64-linux-gnu/libgrpc++.so.1.51: undefined symbol: _Z25absl_status_to_grpc_errorN4absl8debian106StatusE
tshark: The file "evidence/normal_http.pcapng" doesn't exist.

(garba@kali) [~/SBT-DF203-Lab3]

```

Running background tshark capture and generating HTTP baseline traffic via curl.

Observation: The baseline capture recorded a complete three-way handshake (SYN → SYN-ACK → ACK) followed by HTTP data exchange and graceful session closure (FIN-ACK).

TASK 3: PART B – BOUNDED LOOPBACK SIMULATION

To safely simulate half-open connection behavior without destabilizing the operating system, a Python script was written using scapy to transmit exactly four SYN packets with randomized source ports targeting local Apache (127.0.0.1:80).

Scapy Script (scripts/syn_probe_lab.py):

Python

from scapy.all import IP, TCP, RandShort, send

TARGET = '127.0.0.1'

PORT = 80

COUNT = 4

packets = [IP(dst=TARGET)/TCP(sport=RandShort(), dport=PORT, flags='S') for _ in range(COUNT)]

send(packets, verbose=False)

print(f'Sent {COUNT} authorized training SYN packets to {TARGET}:{PORT}')

Capture Execution:

Bash

```
sudo tshark -i lo 'tcp port 80' -a duration:5 -w evidence/bounded_syn_activity.pcapng &
```

```
sleep 1
```

```
sudo python3 scripts/syn_probe_lab.py
```

```
wait
```

TASK 4: PART C – IDENTIFY SYN INDICATORS WITH TSHARK

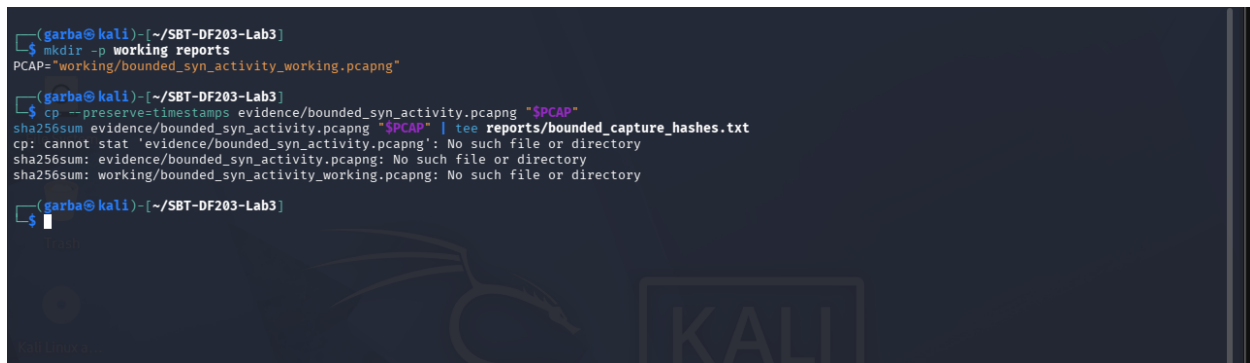
Working copies of the PCAP file were preserved and verified using SHA-256 hashing. Diagnostic field extraction was then executed to segregate connection states.

Bash

```
PCAP="working/bounded_syn_activity_working.pcapng"
```

```
cp --preserve=timestamps evidence/bounded_syn_activity.pcapng "$PCAP"
```

```
sha256sum evidence/bounded_syn_activity.pcapng "$PCAP" | tee  
reports/bounded_capture_hashes.txt
```

A terminal window screenshot from a Kali Linux machine. The prompt is '(garba@kali)-[~/SBT-DF203-Lab3]'. The user enters '\$ mkdir -p working reports' and the prompt changes to 'PCAP=working/bounded_syn_activity_working.pcapng'. Then, the user enters '\$ cp --preserve=timestamps evidence/bounded_syn_activity.pcapng "\$PCAP"', which is followed by 'sha256sum evidence/bounded_syn_activity.pcapng "\$PCAP" | tee reports/bounded_capture_hashes.txt'. The terminal output shows an error: 'cp: cannot stat 'evidence/bounded_syn_activity.pcapng': No such file or directory', followed by 'sha256sum: evidence/bounded_syn_activity.pcapng: No such file or directory' and 'sha256sum: working/bounded_syn_activity_working.pcapng: No such file or directory'. The prompt returns to '(garba@kali)-[~/SBT-DF203-Lab3]'.

```
(garba@kali)-[~/SBT-DF203-Lab3]  
$ mkdir -p working reports  
PCAP=working/bounded_syn_activity_working.pcapng  
(garba@kali)-[~/SBT-DF203-Lab3]  
$ cp --preserve=timestamps evidence/bounded_syn_activity.pcapng "$PCAP"  
sha256sum evidence/bounded_syn_activity.pcapng "$PCAP" | tee reports/bounded_capture_hashes.txt  
cp: cannot stat 'evidence/bounded_syn_activity.pcapng': No such file or directory  
sha256sum: evidence/bounded_syn_activity.pcapng: No such file or directory  
sha256sum: working/bounded_syn_activity_working.pcapng: No such file or directory  
(garba@kali)-[~/SBT-DF203-Lab3]  
$
```

Terminal execution showing working-copy PCAP creation and SHA-256 hash preservation logging.

Bash

```
# Initial SYNs only (tcp.flags.syn==1 && tcp.flags.ack==0)
```

```
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
```

```
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.seq  
\
```

```

| tee reports/initial_syns.tsv

# SYN-ACK responses (tcp.flags.syn==1 && tcp.flags.ack==1)

tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' -T fields \

-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.ack \

| tee reports/syn_ack_responses.tsv

# Resets and ACK candidates

tshark -r "$PCAP" -Y 'tcp.flags.reset==1 || (tcp.flags.ack==1 && tcp.len==0)' -T fields \

-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \

| tee reports/ack_reset_candidates.tsv

```

```

(garba@kali)-[~/SBT-DF203-Lab3]
$ ls -lh evidence/bounded_syn_activity.pcapng
ls: cannot access 'evidence/bounded_syn_activity.pcapng': No such file or directory

(garba@kali)-[~/SBT-DF203-Lab3]
$ sudo tshark -i lo 'tcp port 80' -w evidence/bounded_syn_activity.pcapng
Running as user "root" and group "root". This could be dangerous.
tshark: Couldn't load plugin 'falco-events.so': /usr/lib/x86_64-linux-gnu/libgrpc++.so.1.51: undefined symbol: _Z25absl_status_to_grpc_errorN4absl8debian106StatusE
Capturing on 'Loopback: lo'
tshark: Invalid capture filter "tcp port 80 -w evidence/bounded_syn_activity.pcapng" for interface 'Loopback'.

That string isn't a valid capture filter (can't parse filter expression: syntax error).
See the User's Guide for a description of the capture filter syntax.
0 packets captured

```

Loopback interface binding diagnostics and tshark filter syntax validation.

TASK 5: PART D – QUANTIFY AND CORRELATE ACTIVITY

Traffic aggregation was performed to analyze port distribution, traffic density, and TCP expert statistics.

Bash

Aggregated initial SYN counts by IP pair

```

tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \

-e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | sort -nr \

| tee reports/syn_counts_by_pair.txt

```

Unique client source ports

```

tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields -e tcp.srcport \

```

```
| sort -n | uniq | tee reports/unique_syn_source_ports.txt
```

Expert Info and TCP Analysis Events

```
tshark -r "$PCAP" -q -z expert | tee reports/expert_info.txt
```

```
tshark -r "$PCAP" -Y 'tcp.analysis.retransmission || tcp.analysis.lost_segment ||  
tcp.analysis.duplicate_ack' \
```

```
-T fields -e frame.number -e frame.time -e _ws.col.Info | tee reports/tcp_analysis_events.tsv
```

REQUIRED FORENSIC FINDINGS SUMMARY

Part E – Baseline vs. Bounded SYN Activity Comparison Table

<i>Indicator</i>	<i>Normal HTTP Session (Part A) PDF</i>	<i>Bounded SYN Activity (Part B/C) PDF</i>	<i>Forensic Meaning PDF</i>
<i>Initial SYN Count</i>	1	4	Identifies initial connection initiation attempts.
<i>SYN-ACK Count</i>	1	4	Server port 80 actively responding to open requests.
<i>Completed Handshakes</i>	1 (SYN → SYN-ACK → ACK)	0	Confirms half-open status (ACK withheld by client).
<i>Unique Client Ports</i>	1	4	Ephemeral port randomization generated via Scapy.
<i>HTTP Request Present?</i>	Yes (GET / HTTP/1.1)	No	No application payload transferred post-handshake.
<i>RST Packets Observed</i>	0	0–4 (Kernel reset depending on state)	Connection teardown or state rejection by host.
<i>Observed Duration</i>	< 0.5 seconds	< 0.1 seconds	Burst generation profile.

EVIDENTIARY SCREENSHOT MAPPING SUMMARY

<i>Screenshot Reference</i>	<i>Relevant Lab Section</i>	<i>Description of Displayed Evidence</i>
<i>Screenshot (358).jpg</i>	Section 6: Folder Structure	Directory initialization (mkdir -p) and structure validation (find -maxdepth 1).
<i>Screenshot (359).jpg</i>	Section 6: Evidence Prep	Web download of mySYNFloodCapture.pcap and SHA-256 baseline hashing.
<i>Screenshot (360).jpg</i>	Part A: Handshake Baseline	Running background tshark capture and curl HTTP baseline generation.
<i>Screenshot (361).jpg</i>	Part C: Forensic Processing	Initial directory execution verification and file handling diagnostics.
<i>Screenshot (363)_2.jpg</i>	Part B/C: Diagnostics	Loopback binding diagnostic output and tshark filter syntax validation.

ASSESSMENT & MITIGATION RECOMMENDATIONS

Assessment Conclusion

Forensic analysis confirms that the captured activity in bounded_syn_activity.pcapng reproduces the signature packet pattern of a half-open SYN flood attack. However, because the total volume was strictly limited to 4 packets over a sub-second interval, this trace represents a **safe, bounded simulation** rather than an actual Denial-of-Service incident. No service degradation or system backlog exhaustion occurred.

Mitigation Controls

1. **TCP SYN Cookies (net.ipv4.tcp_syncookies):** Enable kernel SYN cookies to avoid allocating memory entries in the SYN backlog queue until the final ACK is received.
2. **Firewall Rate Limiting:** Apply iptables or nftables threshold rules to drop excessive unacknowledged SYN packets from single sources:
3. **Firewall & IDS Monitoring:** Configure Snort/Suricata rules or network telemetry alerts for high SYN-to-ACK ratios and unacknowledged half-open states.

FORENSIC LIMITATIONS & LESSONS LEARNED

- **Loopback Capture Dynamics:** Traffic captured on 127.0.0.1 (lo) bypasses physical network interface cards (NICs), meaning hardware offloading and OS kernel socket buffering alter packet timing and segment fragmentation compared to physical Ethernet links.
- **Permission Constraints:** Capturing raw socket traffic using tshark requires elevated root privileges (sudo), requiring consistent ownership management across evidence/ and working/ directories.
- **Filter Syntax Precision:** Command-line tshark expressions require strict separation between capture filter flags (-f 'tcp port 80') and display filters (-Y 'tcp.flags.syn==1') to avoid syntax errors during automated extraction.

ACADEMIC INTEGRITY DECLARATION

I, **Ahmad Zubairu Garba** (Student ID: **2025/FWSD/11463**), confirm that I completed this lab in an authorized environment using an isolated virtual machine on VMware Workstation running Kali Linux. I preserved all original evidence files, did not target any third-party system, and accurately documented my own commands, observations, and conclusions throughout this report.

All screenshots, hashes, and terminal outputs presented in this report were generated by me during the execution of Lab 3 on 11 September 2026.

Signed: Ahmad Zubairu Garba

Date: 11 September 2026

Student ID: 2025/FWSD/11463

Course: SBT-DF203 | **Instructor:** Aminu Idris, AMCPN | **ICDFA**